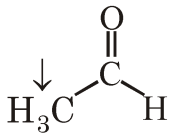
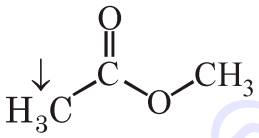
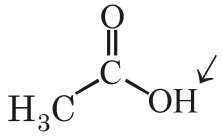
SECTION–I : (i) Only One option correct Type

This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

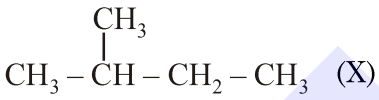
1. The relative acidity of the indicated H in each of the following is

(i) 

(ii) 

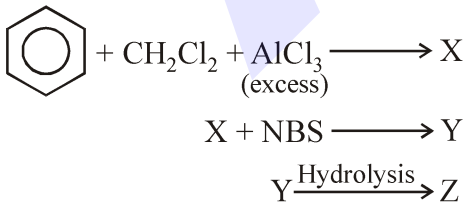
(iii) 

(A) i > ii > iii
(B) ii > iii > i
(C) i > iii > ii
(D) iii > i > ii

2.  (X)

when X undergoes monochlorination and then the products fromed undergo fractional distillation, the number of fractions obtained is :

(A) 1
(B) 2
(C) 3
(D) 4

3. 

Correct option regarding X, Y, Z.

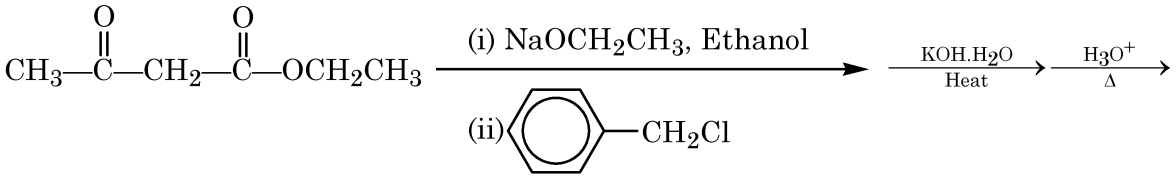
(A) Z when reacts with lucas reagent, white turbidity is formed within 5-10 minutes.

(B) Maximum number of benzonoid structures X can have is two.

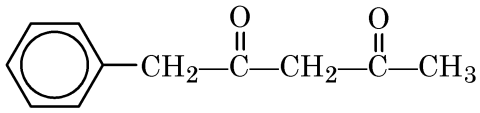
(C) Z when reacts with PCC, benzophenone is formed as product.

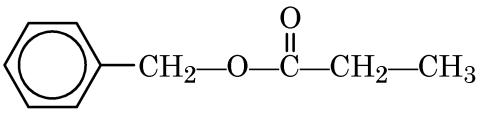
(D) Y when reacts with silver nitrate, white precipitate is formed instant.

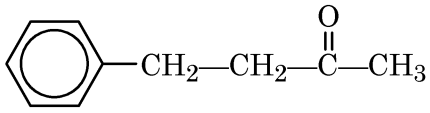
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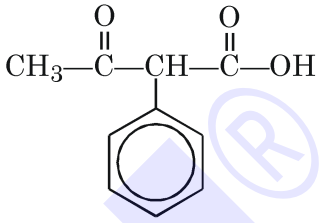
4. 

Which of the following is the final product of the series of reactions shwon above :

(A) 

(B) 

(C) 

(D) 

5. Which of the following is natural condensation Co–polymer ?

(A) Natural rubber (B) Starch

(C) Insulin (D) Dacron

(ii) One or more options correct Type

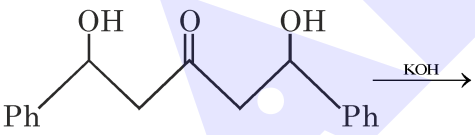
This section contains **05 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

6. Diethyl ether does not react with

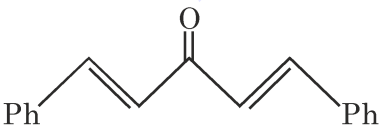
(A) Na (B) $NaNH_2$

(C) Aqueous HCl (D) Aqueous NaOH

7. Consider the reaction given below :



Select the correct statement(s)

(A)  having E, E configuration about double bonds is produced as the major product

(B) The reaction involves $E1_{CB}$ mechanism

(C) Since the intermediate anion is planar, the stereochemistry of the starting compound is irrelevant and the less hindered product is formed

(D) Aldol condensation takes place forming major product

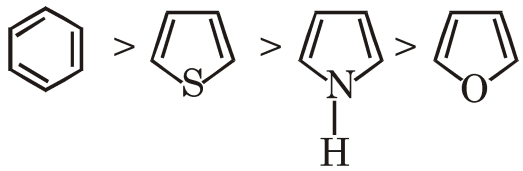
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8. Which one of the following is an oxidation product of a primary amines ?

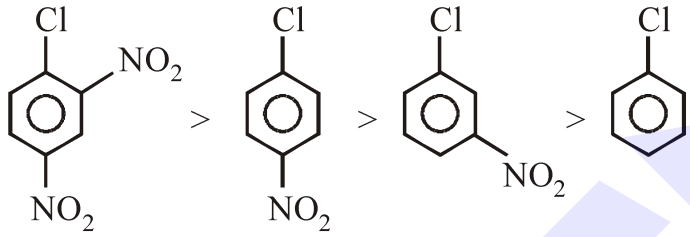
(A) A hydroxylamine (B) A nitroproduct

(C) A nitroso product (D) A alkyl hydrazine

9. Predict correct orders among the following ?

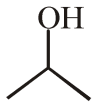
(A)  (resonance energy).

(B) β-D-glucopyranose > α–D–glycopyranose > open chain form of D-glucose (stability).

(C)  (Aromatic Nucleophilic substitution).

(D) $\dot{C}F_3 > \dot{C}C\ell_3$ (% p character in orbital having unpaired electron).

10. Which of the following compounds give(s) iodoform test ?

(A) 

(B) CH_3COCH_3

(C) $CH_3COCH_2COOC_2H_5$

(D) $CH_3-CH(OH)-CH_2-CH(OH)CH_3$

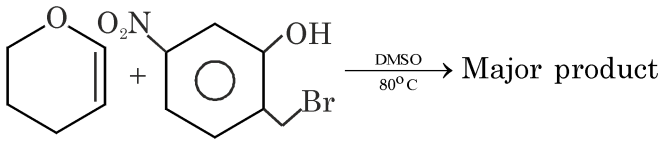
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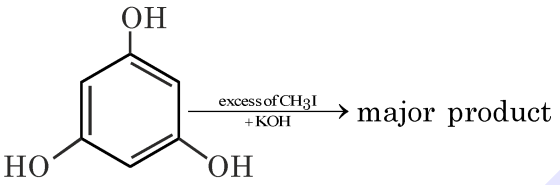
(iii) Paragraph Type

This section contains **01 paragraph** each describing theory, experiment, data etc. **two questions** relate to one paragraph with two questions on each paragraph. Each question of a paragraph has **one or more than one correct answer** among the four choices (A), (B), (C) and (D).

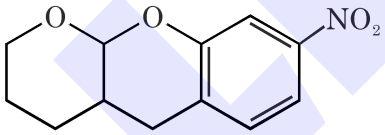
Paragraph for Questions 11 and 12

Consider the scheme of reactions given below :

(I) 

(II) 

11. Select the correct statement(s) regarding reaction (I)

(A) The major product is 

(B) It involves nuclophilic substition

(C) DMSO acts as base

(D) It involves electrophilic aromatic substitution

12. Select the correct statement(s) regarding reaction (II)

(A) The formation of major product involves nucleophic substitution

(B) It involves acid-base reaction

(C) The major product has six methyl groups

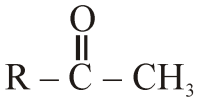
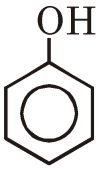
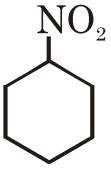
(D) The major product is triketo compound

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(iv) Matching List Type

This Section contains **1 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

13.

	List-I (compounds)		**List-II (tests)**
(P)	$CH_3CH_2CH_2Cl$	(1)	Br_2/H_2O
(Q)		(2)	Baker-mulliken test
(R)		(3)	Iodoform test
(S)		(4)	Beilstein test

Codes :

	(P)	**(Q)**	**(R)**	**(S)**
(A)	4	3	1	2
(B)	2	3	1	4
(C)	3	2	1	4
(D)	2	4	1	3

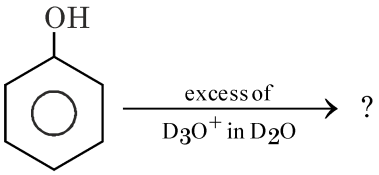
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SECTION-III : (Integer Value Correct Type)

This section contains **05 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

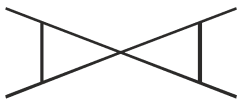
1. All the compounds which gives positive test with Brady's reagent (2,4–DNP) and are having molecular mass 100 are treated separately with $NaBH_4$. Product of how many of these can be used effectively to separate racemic mixture of 2-methylbutanoic acid.

2. Consider the reaction given below

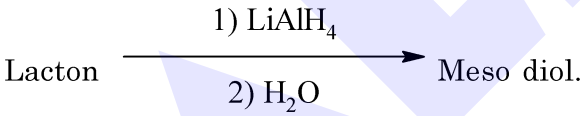


How many hydrogen atoms are replaced by deuterium atoms in the major product ?

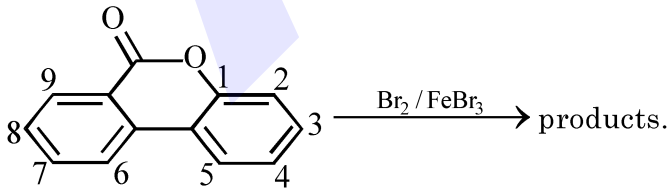
3. Find the difference between the number of optically active and optically inactive isomers of



4. Find all possible **Lactones with six membered ring** having molecular formula **($C_6H_{10}O_2$)**, which when reacts with (1) $LiAlH_4$ (2) H_2O gives meso diol.



5.



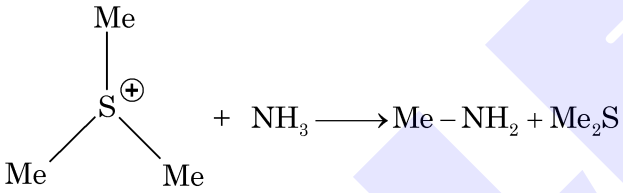
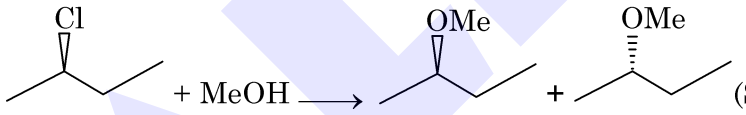
major organic product will have Bromine at position.

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SECTION–IV : Matrix-Match Type

This Section contains **01 question**. Question has **four statements** (A, B, C and D) given in **Column I** and five statements (P, Q, R, S and T) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1.

Column-I	**Column-II**
(A) $CH_3-Cl+NaOH\longrightarrow CH_3OH+NaCl$	(P) Rate of reaction increases in polar protic solvent
(B) $CH_3-Cl+NH_3\longrightarrow CH_3-NH_2+HCl$	(Q) Rate of reaction increases in polar aprotic solvent
(C) 	(R) S_N2 mechanism
(D) 	(S) S_N1 mechanism
	(T) TS has positive charge

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